Search-round report: IDEA-20261023-f15270, re-scope of IDEA-20260803-ff7415

Date 2026-10-23. Task TASK-20261023-4c12dc, batch BATCH-7ca353, goal GOAL-PATH-001, question RQ-MONO-001. Role: idea-generator, policy research-deep. This is a proposal-stage document. **Zero runs, no code, no compute: no number below is measured by this session.** Every statement carries one label.

- **[D]** derived in this report by the idea-generator from cited committed premises. The derivation is written out and is not independently re-derived. It is a derivation note at level 2 of the refutation artifacts in `docs/claims-and-verification.md`, never "proved".
- **[P]** proposed: a conjecture, a design choice, or a transfer assumption.
- **[M]** measured: a number committed in a named record, cited by record and key.
- **[R]** read in this session from a retrieved or frozen source, with its location.

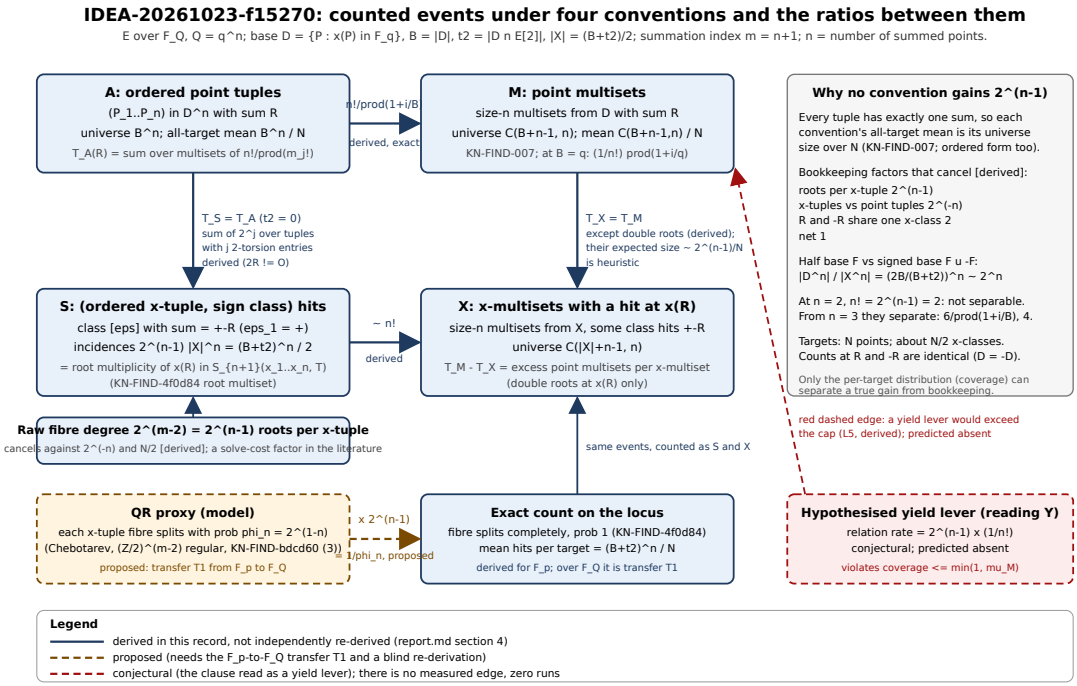


Figure 1: Convention map

Figure 1. Counted events under four conventions and the ratios between them. Solid edges are derived [D]; dashed edges are proposed or conjectural [P]. No edge is measured.

1. The question

`KN-OPEN-62c870` Q3 carries CS-D2 of `DEC-20261017-435b01`: the transfer of the "compounding" clause from F_p to F_{q^n} is "an untested extrapolation and no exponent statement is made". The clause comes from the superseded `KN-FIND-c41ea9` and is quoted here only as superseded text, as `IDEA-20260803-ff7415`, `DEC-20260802-a51c82` and two goal heads carry it: a per-fibre factor of 2 applied once per summation fibre in a Gaudry/Diem-style decomposition over F_{q^n} with n growing compounds as 2^{n-1} = q^{Theta(1)} and WOULD move an exponent.

Decision question. Can the clause be specified as a discriminating hypothesis on a current anchor, which metric carries the discriminating power, and can a derivation alone reach the disposition? (This unblocks `/design-experiment`: DEC-20261017-12e0f4 S-2 readiness (a) and (b).)

Answer in one paragraph. [D] The clause has three readings. Reading Y, a gain in the exact relation rate, is decided by arithmetic: the rate cannot exceed `C(B+n-1,n)/N`, which at `B = q`, `N = q^n` is the Gaudry-Diem `1/n!` itself. Reading P, a correction to a quasirandom proxy, is true and is explained by the regular action of `(Z/2)^{m-2}`: the factor is `1/phi_n = 2^{n-1}`, and it lifts the proxy exactly to the cap and no further. Reading C, a solve-cost factor, is not decided. The mean over all targets is an identity under every convention, so ff7415's conservation band and the all-target trend slope have no power against the clause. The metrics that carry power are the coverage ratio against a matched random base, the generic complete-split frequency `phi_n` (a positive control with slope 1), and the exact per-cell convention ratios.

2. Corpus screen and sources

Paths searched or read (no shell: no `search_knowledge`, no `tools/*` run).

Path	How read	Used for
`ledger/handoffs/TASK-20261023-4c12dc.yaml`	in full	constraints IG-1 to IG-15; sha256 pin **not verified** (no shell)
`ledger/proposals/IDEA-20260803-ff7415.yaml`	in full	carry map
`knowledge/open-problems/KN-OPEN-62c870.md`	in full	Q3 anchor, PF-1, CS-D2, CS-R7
`knowledge/findings/KN-FIND-007.md`, `KN-FIND-4f0d84.md`, `KN-FIND-bdcd60.md`	in full	conservation, root multiset, monodromy
`ledger/decisions/DEC-20261023-19c53b.yaml`	in full	C-Q3 assessment, prior 0.85
`ledger/decisions/DEC-20261017-12e0f4.yaml`	grep, lines 208-297	S-2 readiness, successor
`ledger/decisions/DEC-20260802-a51c82.yaml`	grep, lines 55-94	relation-rate formula, scope clause
`ledger/corrections/CORR-20260802-1d8384.yaml`	in full	the five OI-1 repairs
`ledger/hypotheses/H-MONO-663fb4.yaml`	lines 1-66, 515-585, grep	(I1), (I2), HEUR-CHR-3, K8
`ledger/evidence/EV-MONO-04d447.yaml`, `EV-MONO-0e7ace.yaml`	grep by key	no ordered/multiset key found
`ledger/evidence/EV-MONO-1ef611.yaml`	grep, lines 335-365	R2 text
`experiments/EXP-MONO-c819ba/execution_report.yaml`	lines 120-140, 268-293 only	F_p both-convention fixture (no run directory opened)
`ledger/goals/GOAL-ICEX-001/goal.yaml` 222-250, `GOAL-PATH-001.yaml` 212-236	read	where the clause is carried
`knowledge/frontiers/ecdlp/` README, 8 index-calculus rows in full, all row titles	read	prior_art, dominated_by

knowledge/ literature/KN-LIT-673219.md, inputs/JOUX-VITSE-2010-157/ paper_fulltext.md	lines 56-66, 236-289, 322-353	relation form, 1/n! baseline
docs/ inventor-protocol.md, agents/ idea-generator.md, templates/ research-records.md, tools/ validate_ledger.py (prior_art only)	read	schema
src/crypto_autoresearcher/ index_calculus/ {decompose,tails}.py, experiments/EXP-SMON-e5cbe6/ code/semaev_cover.py	headers only	reuse list

Web (retrieved): two searches; the HAL record `inria-00337631` was refused by an access interstitial; the later version of Joux-Vitse at `https://www-fourier.univ-grenoble-alpes.fr/~viva/research/articles/DLFq5.pdf` was read on pages 1-7. [R] Gaudry (2009) and Diem (2011) themselves were **not read** they are recalled pointers and nothing here rests on them.

Finding of the screen [R]. The frozen Joux-Vitse text states the relation form `R = +-P_1 +- ... +- P_n` with `n` the extension degree (lines 249-256), the equation `f_{n+1}(x_P1,...,x_Pn, x_R) = 0` (line 274), and "the probability of finding one decomposition of a point R is approximately #(F u -F)^n/S_n / #E(F_{q^n}) q^n/n! . 1/q^n = 1/n!" (lines 329-349). The later version (page 5) says "the cost of sign-finding can be reduced by a factor of 2 by fixing the sign of P_1 to + and considering the 2^{n-1} possible signs": in the literature the 2^{n-1} is a cost remark, not a probability.

A correction to ff7415 found by the screen [R]. ff7415's mechanism (1) says `DEC-20260802-a51c82` is "silent on whether the factor compounds". Its text (`scope_that_must_travel_with_it`, lines 80-84) asserts the opposite: the factor "compounds as 2^{n-1} = q^{Theta(1)} and DOES move an exponent". The goal heads say "would" (`GOAL-ICEX-001/goal.yaml` lines 229 and 244; `GOAL-PATH-001.yaml` line 223). The carry map records this as a change with its reason.

3. Re-anchoring and the index map

Index map [D]. A decomposition of `R` into `n` factor-base points, `R = +-P_1 +- ... +- P_n`, holds iff `S_{n+1}(x_1,...,x_n,x(R)) = 0` (Semaev's defining property, `KR-IC-5a43e5`; Joux-Vitse Proposition 2 and equation (4) [R]). So `m = n + 1`. `KN-FIND-4f0d84` gives the fibre `S_m(x_1..x_{m-1},T)` degree `2^{m-2}` over sign classes with `eps_1 = +1`; with `m-1 = n` points there are `2^n` sign vectors and `2^{n-1}` classes modulo the global sign. The expected `m = n+1` is confirmed. **Consequence [D]:** `k = m - 1` is the number of summed points in any field. A `k`-point decomposition in F_p uses the same `S_{k+1}` and the same fibre of `2^{k-1}` classes. The clause's "2^{n-1}" is a function of the point count; the extension degree enters only because Gaudry-Diem take `k = n`. This also explains why `DEC-20260802-a51c82` can say "constant at fixed m": `m = 3` there is `k = 2` and the factor is `2^1 = 2`.

Known-answer control restated on KN-FIND-4f0d84 [P for F_Q]. On the factor-base locus `S_{n+1}(x_1..x_n,T) = c prod (T - x(sum eps_i P_i))` over sign classes with the sum not `0`, splitting completely, degree `2^{n-1}` minus the classes summing to `0`. The committed scope is F_p, toy tier, derivation (corroborated at `m = 3, 4` exhaustively on two curves and `m = 5` at random with zero exceptions, CS-QB [M]). Over F_Q this is transfer **T1**, not an established fact. "193/193" is the old figure from the superseded entry and appears here only as history.

4. Derivations (all [D] unless stated)

Notation. `Q = q^n`. `E: y^2 = f(x)` nonsingular over `F_Q`, characteristic above 3, `N = |E(F_Q)|`. `D = {P in E(F_Q) \ {O} : x(P) in F_q}`, so `D = -D`. `B = |D|`. `t2 = |D n E[2]|` (at most 3). `X = x(D)`, `|X| = (B+t2)/2`. Counts at target `R`:

- `T_A(R)`: ordered tuples `(P_1..P_n)` in `D^n` with sum `R`;
- `T_M(R)`: size-`n` multisets from `D` with sum `R`;
- `T_S(R)`: pairs (ordered x-tuple `xi`, class `[eps]`) with `sum eps_i l_i = +-R`, for fixed lifts `l_i` of `x_i`; for `R != O` this is the multiplicity of `x(R)` as a root of `S_{n+1}(xi, T)`;
- `T_X(R)`: size-`n` multisets of `X` having at least one class with sum `+-R`.

4.1 Conservation under each convention

L1 [D]. Every ordered tuple has exactly one sum, so `sum_R T_A(R) = B^n` and the all-target mean is `B^n/N`. (`H-MONO-663fb4` (I1) states this over F_p; the proof is the same for any finite abelian group.) `KN-FIND-007` gives `sum_R T_M(R) = C(B+n-1,n)`.

L2 [D], convention ratio of means. `mean T_A / mean T_M = B^n / C(B+n-1,n) = n! / prod_{i=0}^{n-1} (1 + i/B)`, which lies between `n! exp(-n(n-1)/(2B))` and `n!`. Per target, a multiset with multiplicities `m_j` has `n!/prod(m_j!)` orderings, so `T_A(R) = sum over multisets mu with sum R of n!/prod(m_j!)`. The fraction of multisets with a repeated point is `1 - prod_{i<n}(B-i)/(B+i)`. This matches the F_p observation in `EXP-MONO-c819ba` (lines 125-134 [M]) that `F^m/N` and the multiset mean "agree only asymptotically", with no single m-independent convention factor.

Nominal values at `B = q` (actual `B` differs from `q` by `O(sqrt q)`):

n	q	A/M mean ratio `n!/prod(1+i/q)`	compare `n!`	compare `2^{n-1}`
2	61	1.968	2	2
3	17	5.070	6	4
3	251	5.929	6	4
4	11	14.63	24	8
4	101	22.63	24	8
5	7	36.38	120	16
5	29	86.42	120	16

At `n = 2` the pair `n!` versus `2^{n-1}` is not separable (both 2); from `n = 3` they separate. At small `q` the ratio sits far below `n!`, so the comparator is always the exact L2 value.

4.2 Baseline reproduction (proof_search_map, audit 1)

With `B = q` and `N = q^n`: `C(q+n-1,n)/q^n = (1/n!) prod_{i=0}^{n-1}(1 + i/q) -> 1/n!`, because `C(q+n-1,n) = q(q+1)...(q+n-1)/n!`. This is the Joux-Vitse `1/n!` [R, lines 329-349] and `KN-FIND-007`'s remark that the Gaudry-Diem probability is the conservation mean. Examples: `n = 3, q = 17`: `(1/6)(18/17)(19/17) = 0.1972` against `1/6 = 0.1667`; `n = 5, q = 29`: `0.01157` against `1/120 = 0.00833`. The baseline is a parameter slice of the family `C(B+n-1,n)/N`, so the proposal is a family extension, not an independent method.

4.3 The sign-class convention

L3 [D]. For `R` with `2R != O`: `T_S(R) = sum over ordered tuples with sum R of 2^{j}`, where `j` counts entries in `E[2]`. Proof. Fix lifts `l_i`. The map `(xi, eps) -> (eps_i l_i)` from `X^n x {+-1}^n` to `D^n` is onto and is `2^j`-to-one over a tuple with `j` two-torsion entries (`l = -l` for those). The sums satisfy `s(-eps) = -s(eps)`. For a class `{eps, -eps}` with `s = +-R`, exactly one of the two vectors sums to `R`, because `R != -R`. So the number of classes with `s = +-R` equals the number of vectors

eps with s(eps) = R, which is the sum above. Hence T_S = T_A whenever t2 = 0, and sum_R T_S ~ (B+t2)^n. The factor 2^{n-1} therefore does **not** appear between S and A.

L4 [D], x-multisets. T_M(R) - T_X(R) = sum over x-multisets eta of (number of point multisets over eta with sum R, minus 1), over those eta with at least one; two distinct multisets over one eta summing to R are two distinct classes with s = +-R, i.e. a double root at x(R). Under a random model the expected relative size is O(2^{n-1}/N) [P, heuristic]; each cell reports the exact count.

Exact known answers at n = 2 [D]. T_A(O) = B (the pairs (P,-P)), T_M(O) = (B+t2)/2 (one multiset per x-class). These are exact, cheap, and test the counter on a degenerate target.

Cancellation-free count [D]. A multiset with a pair {P,-P} (or {P,P} for a 2-torsion P) sums into an (n-2)-fold sum. The number of multisets with no such pair is the coefficient of z^n in ((1+z)/(1-z))^{(B-t2)/2} (1+z)^{t2}. Check at n = 2, t2 = 0, b = B/2: coefficient 2b^2, equal to C(B+1,2) - b = b(2b+1) - b. Generic targets exclude E[2], O and the sums of n-2l base points; their mean is (N_cf - m_S)/(N - |S|) with m_S of relative size ~ q^-2.

4.4 Where 2^{n-1} is real: the proxy factor

L6 [D given T1]. By KN-FIND-bdcd60 (3), for x_1..x_{m-1} in the field with every f(x_i) != 0 and the fibre squarefree of full degree, Frobenius acts on the 2^{m-2} roots as translation by the class of (chi(f(x_1)),...,chi(f(x_{m-1}))) modulo +-(1,...,1), and the cycle type is 1^{2^{m-2}} iff all chi(f(x_i)) are equal. With m - 1 = n independent signs, phi_n = P(all equal) = 2 . 2^{-n} = 2^{1-n} (up to O(n/sqrt Q) from Weil). This equals 1/|G| for G = (Z/2)^{m-2} acting regularly (KN-FIND-bdcd60 (1)): |G| = deg = number of sign classes. That explains the coincidence ff7415 noticed. On the factor-base locus every chi = +1, so the split probability is exactly 1 (KN-FIND-4f0d84).

Quasirandom proxy QR [P]. Treat each ordered x-tuple's fibre as splitting completely with probability phi_n, and, if it splits, its 2^{n-1} roots as uniform on the ~N/2 x-classes. Mean hits per target: |X|^n phi_n 2^{n-1} (2/N) = 2|X|^n/N. The exact value is 2^{n-1}|X|^n (2/N) = (B+t2)^n/N. **Exact over proxy = 1/phi_n = 2^{n-1}.** At n = 2 this is the committed 1/freq_split ~ 2 of EV-MONO-a0a89c OBS-4 (as EV-MONO-1ef611 R2 records it [M]). The factor corrects a proxy, as ff7415 conjectured, and lifts the proxy exactly to the conservation mean, no further.

L7 [P]. The DEC-20260802-a51c82 formula (1-1/W_eff)/freq_split -> 2(1-1/W_eff) is the m = 3 slice. The general-m form would be 2^{m-2} prod_{i=1}^{m-2}(1 - i/W_eff), reducing to the committed formula at m = 3. Not derived; stated so the design step checks it against the F_p fixture. If it holds, "2" is not the limit at m >= 4 in prime fields, independent of extension fields.

4.5 The cap (reading Y)

L5 [D]. Let T = T_M(R) for R uniform over the group, mu = E[T] = C(B+n-1,n)/N, and c = P(T >= 1) the coverage (relation rate per target). Since 1{T>=1} <= T and c <= 1: c <= min(1, mu). By Cauchy-Schwarz, mu = E[T 1{T>=1}] <= sqrt(E[T^2] c), so c >= mu^2/E[T^2]. For any base of size B, the relation rate is at most C(B+n-1,n)/N: at B = q this is (1/n!) prod(1+i/q). **No base, structured or random, and no per-fibre factor can lift the rate above the Gaudry-Diem 1/n! (times that product).** A claim of 2^{n-1}/n! exceeds the cap whenever n(n-1)/(2q) < (n-1) ln 2, i.e. for n < 1.38 q. KN-FIND-007's 1.582 is the matched-random version: gain <= min(1,mu)/(1 - e^{-mu}).

Poisson-null ceilings for the coverage ratio against a matched random base (mu = 1/n!, [D] arithmetic):

n	mu = 1/n!	1 - e^{-mu}	cap ratio mu/(1-e^{-mu})	log2 ceiling
2	0.5	0.3935	1.2707	0.346

3	0.1667	0.1535	1.0857	0.119
4	0.04167	0.04081	1.0210	0.030
5	0.008333	0.008299	1.0042	0.006

The ceilings fall with `n`. A fitted slope of `log2 kappa_rel` against `n-1` is therefore bounded above by a negative number under the null, so F1 (slope `>= 0.5`) is unreachable unless a premise fails. The Poisson null on the coverage-over-cap ratio `(1-e^{-mu})/mu` has `log2` values `-0.346, -0.119, -0.030, -0.006` at `n = 2..5` and a least-squares slope of `+0.111` against `n-1`: the null alone uses 74 percent of the band `[-0.15, 0.15]`. That is why the fitted object is the line-to-matched-random ratio, which cancels this trend.

4.6 Quantifier order (proof_search_map, audit 3)

L8 [D]. `2^{n-1} = q^{Theta(1)}` forces `n = Theta(log q)`: `q` depends on `n`. With `n = c log2 q`, `n! = q^{(c/ln 2)(ln n - 1)(1+o(1))}`, superpolynomial, while `2^{n-1} = q^c / 2`. A genuine gain `2^{n-1}` in the rate `1/n!` would change `log_q(1/rate)` by the relative amount `ln 2/(ln n - 1) -> 0`: not an exponent-class change. Against the published bound, the Gaudry-Diem hidden constant is `2^{O(n^2)}` (`KN-LIT-673219` lines 63-66 [R]; `n! 2^{n(n-1)}` in `KR-IC-46d822`), so a `q^c` factor is lower order next to a `q^{Theta(log q)}` upper bound. This reads relative to an upper bound and says nothing about the true cost. A fixed `q` ladder with `n <= 5` never enters the regime and decides only the premises. The "for all n" content of the conclusion comes from the closed forms L1-L5, not from the grid.

5. What a derivation reaches, and what the measurement adds (IG-9, IG-12)

The corpus already decides part of it, plainly. `KN-FIND-007` Consequence 2 states that "any growth with N clause on mean yield is identically satisfied at slope zero", and its "What this does not say" bullet states that the Gaudry-Diem `1/n!` is the conservation mean. Both are committed (`confidence: established`, derivation, toy). What the corpus does **not** hold is (i) the origin of `2^{n-1}` (L6), (ii) the cap written against the clause (L5), (iii) the four-convention map (L2-L4), and (iv) the transfer T1.

Derivation alone reaches [D, pending independent re-derivation]: reading Y and the mean (L1-L5), the convention table (L2-L4), the origin of `2^{n-1}` given T1 (L6), and the non-exponent-class statement (L8). The three named dispositions (`CLAUSE_REFUTED_AT_TOY`, `CONVENTION_ARTIFACT_DIAGNOSED`, `CLAUSE_SUPPORTED`) need the measurement by their names ("at toy") and are assigned by the Coordinator, not by this record.

What the measurement adds that the derivation does not: 1. T1: the F_p derivations (4f0d84, bdcd60) over `F_Q`, at the tested `(q, n)`. 2. Counter correctness, through exact known answers and a failing twin. 3. The Poisson per-target law and coverage against `1 - e^{-lambda}` (HEUR-F-1, F-2). 4. The size of the small-`q` corrections (repeats, 2-torsion, cancelling pairs). 5. A positive control that the toolchain can see a `2^{n-1}` trend where one exists (PR-5).

It cannot add anything about `n` beyond the tested ladder, or about reading C.

Cheapest artifact that would make the derivation official. A zero-run, blind re-derivation review of L1-L6 (the pattern used for `KN-FIND-4f0d84` and `KN-FIND-bdcd60`), followed by a knowledge entry for `KN-OPEN-62c870` Q3 that supersedes the "WOULD move an exponent" reading. The n = 2 and n = 3 exhaustive cells (enumeration sizes below) corroborate it in a single short run. Conditional parts (L6 given T1; coverage statements given HEUR-F-1, F-2) are to be recorded under the two-record rule of `docs/claims-and-verification.md`.

6. Design (IG-5, IG-7)

6.1 Exhaustive grid, arm A

Group `E(F_Q)`, order `N` found by exhaustive count. Factor base `D = {P : x(P) in F_q}`, `B ~ q`. Curves per `(n, q)`: 3, coefficients `(a,b)` uniform over `F_Q` from a seeded generator, reject singular, reject `j(E) in F_q` (this also removes `j = 0, 1728`). Field `F_Q = F_q[t]/(g)`, `g` the least monic irreducible of degree `n`. Identity fields to freeze: `q, n, g, a, b, N, t2, B, |X|, seed`. Targets: all `N` points; generic targets exclude `E[2]`, `O` and sums of `n-2l` base points.

n	q ladder	`Q = q^n`	multisets `C(q+n-1,n)` at `B = q`
1 (anchor)	61, 251	61, 251	all ratios must be 1
2	61, 251, 1021, 4093	3,721; 63,001; 1,042,441; 16,752,649	1,891; 31,626; 521,731; 8,378,371
3	17, 41, 101, 251	4,913; 68,921; 1,030,301; 15,813,251	969; 12,341; 176,851; 2,667,126
4	11, 23, 53, 101	14,641; 279,841; 7,890,481; 104,060,401	1,001; 14,950; 367,290; 4,598,126
5	7, 11, 17, 29	16,807; 161,051; 1,419,857; 20,511,149	462; 3,003; 20,349; 237,336

Counts are derived from the parameters, not timings. Ordered x-tuple incidences are `(B+t2)^n/2`, about `Q/2`. All 48 grid cells (4 `n` x 4 `q` x 3 curves) and the 6 anchors are exhaustive over all targets; none is sampled. Per curve three bases are run: the line base, a matched random symmetric base (null), and a half-size random base (failing twin). Cells with `B < n^2` are flagged repeat-dominated; fits are reported with and without them. The largest `Q` is `104,060,401 < 2^27`, toy tier by the mechanical rule (field size at most 32 bits).

6.2 Arms B and D

Arm B. Per curve per `n`, `10^6` uniform ordered x-tuples in `F_Q^n`: complete-split frequency `phi_hat_n` against `2^{1-n}`. Standard errors: `0.0005` at `n = 2` (`phi = 0.5`), `0.00024` at `n = 5` (`phi = 0.0625`). `S_{n+1}` is evaluated numerically by the resultant recursion, cross-checked by the group-law method on a subsample.

Arm D (optional). Prime-field window base, `k = 2, 3, 4` points (`m = 3, 4, 5`), same convention table, to show that the convention ratios and the factor `2^{k-1}` do not depend on the field degree. Reuses `EXP-MONO-c819ba` outputs as a fixture; not re-measured.

6.3 Sampled validation arm C (the heuristic-validation arm)

Required by `DEC-20261017-12e0f4` successor_after_this_lane, `agents/coordinator.md` "Dispatch bias" and `docs/target-result-profile.md` C12. **Design specification only.**

- **Method: target-sampled, base-exhaustive.** Enumerate all `C(B+n-1,n)` multisets once, sum each, and look the sum up in a hash set of `s` sampled target classes (`R` and `-R` share a class; the counts agree because `D = -D`). No table of size `N`. Per-target counts are exact; the only sampling is the targets. `N` by an SEA/Schoof point counter.
- **Scale.** `Q` between `2^30` and `2^32` (toy by the mechanical rule): `n = 2`, `q` about `2^15` to `2^16`; `n = 3`, `q` about `2^10`; `n = 4`, `q` about `2^8`; `n = 5`, `q` about `2^6`; `n = 6`, `q = 37`; `n = 7`, `q = 23`. Multisets per cell about `Q/n!` (at `Q = 2^32`: `2.1e9`, `7.2e8`, `1.8e8`, `3.6e7` for `n = 2..5`; `C(42,6) = 5,245,786` for `n = 6, q = 37`; `C(29,7) = 1,560,780` for `n = 7, q = 23`). An optional extension to `Q` about `2^36` (`n = 6, q = 61`; `n = 7, q = 31`) is **medium tier** by the mechanical rule (`32 < bits <= 96`),

still far from cryptographic scale. `n = 6, 7` sit at `B/n^2` about `1.0` and `0.47`: repeat-dominated, handled exactly by L2.

- **Instances and seeds.** Curve seed = SHA-256 of the string `f15270|n|q|index`; the PRNG is named by the design step; 3 curves per `(n, q)`; targets by uniform `x in F_Q` with `chi = +1` and a uniform sign, one representative per class; a null base and a twin base per curve; every seed in the contract.
- **Random model with parameter tied to the conservation mean.** On generic targets `T_M(R) ~ Poisson(lambda), lambda = (N_cf - m_S)/(N - |S|)`, with `N_cf` from the generating function of section 4.3. `lambda` is the conservation mean of L1-L2 corrected for cancelling pairs; there is no free parameter. [P: HEUR-F-1.]
- **Sample sizes and power.** `s >= (1-c)/(c eps^2), c = 1 - e^{-lambda}, eps = 0.03` relative SE on coverage. At nominal `lambda = 1/n!` (an upper bound on `s`, since the actual `lambda` is larger at small `B`): `n = 2`: 1.7e3; 3: 6.1e3; 4: 2.6e4; 5: 1.3e5; 6: 8.0e5; 7: 5.6e6 [D arithmetic]. A coverage ratio of `1.10` against `1` is then `3.3` SE; the clause's `2^{n-1}` and the `1.582` cap are many SE away.
- **Machine-readable CDF output.** One JSON file per cell, `results/<cell_id>.cdf.json`: `cell_id, n, q, curve_id, seed, B, t2, N, lambda, s, hist, cdf_empirical, cdf_poisson, ks_stat, chi2_stat, chi2_df, zero_fraction, coverage, dispersion_index, max_count, p_max_pred, exceptional_targets`.
- **Tail-consistency checks.** (T1) the observed maximum count against the Poisson order statistic, `P_pred(max >= observed) >= 0.001`; (T2) the fraction of targets with count `>= 3`; (T3) the zero fraction against `e^{-lambda}`; (T4) a structural tail: counts at `R in S` exceed Poisson by construction (for `n = 3` and `R in D`, `T_M(R) >= |X|`), reported separately as a counter-sensitivity check.
- **Transfer assumption, stated.** Poisson at the tested `(q, n)` is indicative at larger `q` for the same `n` (`B/n^2` grows). The derivation L1-L5 needs no such transfer.

7. Pre-registration, with each metric's power (IG-3)

Carried from ff7415: the conservation band `[1/1.582, 1.582]`; the trend-slope band `[-0.15, 0.15]` against a compounding prediction of `1.0`; F1 at slope `>= 0.5`; F2 to F5; the three dispositions. Nothing is loosened and no data exist to loosen it against.

Metric	Vacuous by KN-FIND-007?	Power
PR-1 conservation ratio over all targets	**yes**, an identity (exactly 1 in an exhaustive cell)	counter and convention defects only; the band passes a 40 percent defect
PR-2a slope of `log2(PR-1)` against `n-1`	**yes**, identically 0	none; kept so its inability to fire is on the record
PR-2b slope of `log2 kappa_rel` (line coverage over matched-random coverage)	no	fires only through a failed premise; ceilings in 4.5 fall with `n`
PR-3 convention ratios (exact, per cell)	partly	detects a counter or encoding error; the n!-versus-2^{n-1} test works from `n = 3`
PR-4 split completeness over `F_Q`	not a yield metric	tests transfer T1
PR-5 generic split frequency `phi_n`, slope of `log2(1/phi_n)`	no	the one real `2^{n-1}`; slope 1.0 in `[0.85, 1.15]`

PR-6 Poisson law, coverage against `1 - e^{-lambda}`	no	tests HEUR-F-1, F-2

Metrics that carry the discriminating power. PR-5 (positive, slope 1) against PR-2b (null, slope 0): the two explanations of the clause, a proxy factor versus a yield factor, give predicted slopes that differ by `1.0`, more than six times the band `0.15`. PR-3 separates a convention artifact (ratio `2^{n-1}` or `2^n`, separable from `n!`) from the tautology `A/M = n!/prod(1+i/B)`.

Repair of ff7415's artifact trigger. ff7415 diagnoses a convention artifact if the ordered count over the multiset count "tracks `2^{n-1}` OR `n!`". By L2 that ratio tracks `n!` whatever the clause is, so the trigger fires on a tautology and has no power. The label now requires a ratio that tracks `2^{n-1}` or `2^n` and is separable from `n!` (`n >= 3`).

What each outcome would mean.

Outcome	Reading
PR-1, 2b, 3 agree; PR-5 slope 1; controls detect	reading Y refuted at the tested cells, the factor is a proxy factor: `CLAUSE_REFUTED_AT_TOY` with `CONVENTION_ARTIFACT_DIAGNOSED` (compatible, not exclusive)
PR-2b slope `>= 0.5` or the cap exceeded at a cell	`CLAUSE_SUPPORTED`: the measured event is not the counted event, or a premise (cap, Poisson null, counter) is false; event identification before any exponent claim
PR-5 not near `2^{1-n}`	the proxy reading P or transfer T1 fails: the origin of `2^{n-1}` is unexplained, but Y stays refuted by L5
a positive control not detected (F6)	no null may be claimed; the family is void as a refutation, the derivation is unaffected
an exact identity L2-L4 fails (F7)	derivation error; correction record

8. Controls (IG-6, declared before belief)

Control	Object	Expected	Power
(a) null base	uniformly random symmetric base of size `B` (`t2` matched where possible)	all-target mean ratio exactly 1; 3 SE if sampled	counter defects
(a2) failing twin	random symmetric base of size `ceil(B/2)`	ratio `C(B/2+n-1,n)/C(B+n-1,n)`, about `2^{-n}`, exact per cell	proves the counter can see a `2^n` effect
(b) nearby object	`n = 1` anchor; prime-field window arm D	factor `2^0`, all ratios 1	confirms no spurious compounding where the clause predicts none
(c) positive PC-A	slope of `log2(A/M)` against `n-1`: derived `1.972` at `B -> infinity`	within 0.15 of the L2 value at the actual `B`	the method must detect a slope above

	(log2 n! = 1, 2.585, 4.585, 6.907)		1 built into the convention
(c) positive PC-B	slope of log2(1/phi_n)	1.0	the one real 2^{n-1}
(c) positive PC-C	the failing-twin ratio	about 2^{-n}	2^n bookkeeping visible
(d) known answers	n = 2: T_A(O) = B, T_M(O) = (B+t2)/2; split completeness on the locus (F_p scope, CS-QB [M])	exact	counter and transfer
(e) exact denominator	every ratio is against an exact rational	no certifying gate to fail	EV-ECDLP-008 lesson

Each control has a failing twin or an exact value, so none can pass without power (OI-1 repair 3).

9. OI-1 repairs, and overlap (IG-10)

PF-1 of KN-OPEN-62c870 applies to any successor census of a relation-search statistic. No waiver is claimed here; a waiver is for the Coordinator. The five repairs of CORR-20260802-1d8384 OI-1:

Repair	Handling in an F_{q^n} decomposition count
R1 reverification-disposition rule (defect 3)	**Applied, adapted.** Every cell has an independent recount (S from ordered x-tuples, M from multisets). A cell whose recount ends in failed_infrastructure is void and never counted for or against; an outcome depending on it is inconclusive.
R2 2-torsion generator guard (defect 7)	**Applied, adapted.** t2 is reported per curve and carried in L3 and N_cf; E[2] targets are excluded from the generic set; j not in F_q removes the j = 0, 1728 automorphism curves.
R3 relative tolerances on window controls (defect 4)	**Applied.** Tolerances are exact equality or z-scores against the predicted SE, never absolute; each control has a failing twin.
R4 FAMILY_RATE_ALERT	**Not applicable as defined.** There is no exceptional-locus family in a decomposition count. Its purpose, multiplicity control, is met by Holm correction across cells and a reported flagged-cell fraction.
R5 uniform (A,B) draw (defect 9)	**Applied.** (a,b) uniform over F_Q, not a small-height box.

Overlap with H-MONO-663fb4. Its statement carries (I1) the ordered mean and (I2) the exact variance over F_p; HEUR-CHR-3 and K8 call the ordered/multiset correspondence a

declared stage and cite ff7415. EV-MONO-04d447 and EV-MONO-0e7ace carry no ordered/multiset key (grep by key, no match); they rest on EXP-MONO-c819ba, whose execution_report.yaml (lines 122-139, 268-282) reports both conventions at every F_p cell and that F^m/N is not KN-FIND-007's mean. **Reused as a fixture, not re-measured:** the F_p per-cell mean_ordered, mean_multiset, var_ordered, var_multiset of EXP-MONO-c819ba raw-result.json, to be read by key by the design step. Raw numbers live in run directories, which this session did not open. **Not covered there:** F_{q^n}, conventions S and X, coverage and the cap, phi_n.

10. Carry map summary (33 entries covering every ff7415 field)

8 kept, **24 changed**, **1 dropped** (the id). The per-field table with reasons is in the proposal's carry_map. The substantive changes:

- Anchor moved from KN-FIND-c41ea9 to KN-OPEN-62c870 Q3 and KN-FIND-007; root-set facts from KN-FIND-4f0d84; "193/193" is history only.
- Mechanism (1) corrected: DEC-20260802-a51c82 asserts the clause, it is not silent.
- Counting widened from two conventions to four (A, M, S, X) with exact predicted ratios.
- The all-target trend slope is declared vacuous; the same band is applied to the coverage ratio against a matched random base.
- The artifact trigger repaired (tautology).
- Added: the sampled validation arm C, positive controls PC-A to PC-C, the failing twin, the proof_search_map, the OI-1 handling.
- Kept: the conservation band, the slope band [-0.15, 0.15] and compounding value 1.0, F1 at 0.5, F2 to F5, the three dispositions, toy tier, "moves no exponent in either direction".

11. Prior and recommendation

Prior, recorded before any work. At least 0.90 that the measurement agrees with the derivation; disagreement paths are a derivation error, a failed transfer T1, a counter defect, or the clause meaning reading C. DEC-20261023-19c53b recorded 0.85 for artifact or refuted.

Recommendation to the Coordinator (not an assignment). Rank the zero-run blind re-derivation of L1-L6 first, since it decides readings Y and the mean at the lowest cost; then / design-experiment with arms A and B as the primary contract, arm C as the paired heuristic-validation arm, and a proof-of-concept implementation task in the same batch. **Cheapest valid discriminator to run first:** the n = 2 and n = 3 exhaustive cells with the null base and the failing twin (enumeration at most 2,667,126 multisets at B = q), because they test the exact identities L2-L4, the known answer T_M(0) = (B+t2)/2 and the counter's power at once.

Blockers for /design-experiment (S-2 readiness). (a) re-anchor: cleared by this record. (b) sampled arm: specified here. (c) implementation: not cleared; it needs F_{q^n} arithmetic for n up to 7, point counting, the exhaustive histogrammer and numeric S_{n+1}. Reusable by path (headers read, bodies not): experiments/EXP-SMON-e5cbe6/code/semaev_cover.py (F_p and F_{p^2}), src/crypto_autoresearcher/index_calculus/decompose.py, tails.py. Also: an independent re-derivation of L1-L6 is advisable before a contract freezes the predicted values.

12. Limitations

- **[D] is not verified.** Nothing here is independently re-derived. A reviewer should treat L1-L8 as a derivation note awaiting a blind re-check.
- **Scope.** Toy tier. The grid is at most Q < 2^27; arm C at most 2^32 (toy) with an optional 2^36 (medium by the mechanical rule). CS-D2 and CS-R7 bind: the transfer from F_p to F_{q^n} is an untested extrapolation and no exponent statement is made. **This moves no exponent in either direction.**
- **T1 is a transfer.** Both 4f0d84 and bdcd60 are F_p statements. L6 and PR-4, PR-5 are conditional on T1. RQ-MONO-001's own scope lists prime fields only; ownership is recorded in DEC-20261023-19c53b.

- **Reading C is untouched**: fibre degree, Groebner cost, sign-finding cost, and a per-attempt success inside one algorithm. The cap bounds relations per target, hence attempts times success, not cost. Larger factor bases (`B > q`), target-dependent bases (`KN-OPEN-020`), relation rank and independence (`KN-FIND-007` "not captured") are untouched.
- **Literature.** Gaudry and Diem were not read. The `1/n!` baseline rests on the Joux-Vitse restatement. Whether L2-L5 are already in the published literature was not checked.
- **Not done (no shell).** The sha256 pin of the card, `tools/allocate_id.py --check`, `tools/validate_ledger.py`, `tools/build_frontier_map.py --match` and `search_knowledge` were not run; the YAML was checked by eye only. The proposal exceeds the 8 KiB advisory cap (an advisory in `validate_ledger.py`, not an error), because the card's required content cannot fit and its write scope has no sibling notes file.

13. Honest accounting (docs/inventor-protocol.md section 5)

- **Objects considered.** (1) the all-target mean count per convention; (2) the per-target count distribution and coverage; (3) the sign-class root multiset of the fibre; (4) the proxy-to-exact ratio `1/phi_n`; (5) a per-attempt success inside an exhaustive `S_3` solve tree (named, not examined). Established families are off-limits as the lens: summation-polynomial decomposition with Groebner solving (`KR-IC-5a43e5`, `46d822`, `081a58`), symmetrised polynomials and torsion (`KR-IC-955fd6`). No object enumeration for the ECDLP exists (`KN-OPEN-019`), so the list is a sketch.
- **Lossy-projection test.** A, M-with-multiplicities and S are changes of coordinates (L2, L3 recover one from another), so their ratios carry combinatorial weights and no new information about yield. The all-target mean is a lossy projection (it forgets the target) and is exact under every convention, hence vacuous for any claim about bases. Only the coverage (per-target indicator) is a lossy projection that separates bases.
- **Operation set and class.** Operation: translation by factor-base elements; index calculus is Class I, partial-action (`IDEA-20260806-c5d183`). No non-translation operation set is proposed; the record is not an attack, so search bias 6 supplies no candidate.
- **Depth of verified structure.** Derivation tier, unreviewed, toy; no measurement.
- **`dominated_by`:** `n/a (no result claimed)`. Frontier rows checked: `KR-IC-46d822`, `64659b`, `5a43e5`, `081a58`, `955fd6`, `5931ee`, `73db3f`, `9e610d`.
- **`sota_delta`:** no attack; measurement/correction contribution only. The clause would put the rate `2^{n-1}` above the `1/n!` baseline (2, 4, 8, 16 times at `n = 2..5`); L5 caps it at a ratio of 1 to the baseline.
- **Closures, with mechanism (inventor-protocol section 4).** Closed: the clause read as a gain in relation rate at fixed factor-base size. Obstruction: exact conservation plus the coverage cap (L1, L5). Argument: every tuple has one sum, so only universe-size ratios differ. Not closed, and naming what remains: reading C (solve cost), larger factor bases, target-dependent bases, relation rank and independence, and any `n` beyond the ladder.
- **Open directions.** (1) blind re-derivation of L1-L6; (2) the general-`m` form L7 against the F_p fixture, which matters to `GOAL-ICEX-001`'s read of `DEC-20260802-a51c82` independently of extension fields; (3) the per-attempt success of `decompose.py`'s exhaustive tree (reading C); (4) a `KN-LIT` entry for the later Joux-Vitse version read here.

14. Citations with provenance

Ref	Provenance	Used for
`KN-LIT-673219` (Joux-Vitse, ePrint 2010/157, frozen)	kb, read by this session at lines 56-66, 236-289, 322-353	relation form, `f_{n+1}`, `1/n!`, constants

https://www-fourier.univ-grenoble-alpes.fr/~viva/research/articles/DLFq5.pdf	retrieved, pages 1-7, not frozen	sign-finding as a cost remark (p.5)
KN-FIND-007, KN-FIND-4f0d84, KN-FIND-bdcd60, KN-OPEN-62c870	internal, read in full	premises
KN-LIT-001, KN-LIT-002, KN-LIT-003 (Semaev, Gaudry, Diem)	recalled as pointers; abstract_read in frontier rows	not relied on
DEC-20260802-a51c82, CORR-20260802-1d8384, EXP-MONO-c819ba, H-MONO-663fb4	internal, read at the stated lines	formula, repairs, fixture, overlap